Kasra **Mossayebi**

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Education

University of Toronto *Toronto, Canada*

BSc in Mathematics *September 2025 -*

- Pursuing mathematics specialist, 3.85 cGPA
- Took Analysis 1 (MAT157), Algebra 1(MAT240), and Algebra 2 (MAT247), Analysis 2(MAT257), Abstract Algebra (MAT347), Advanced Ordinary Differential Equations(MAT267), and Topology (MAT327)
- Member of "∞—Categories in Algebraic Geometry" directed reading group

Northeastern University *Boston, United States*

Visiting Student Researcher *Summer 2026*

- Official visiting student at Northeastern doing research with the Halverson lab
- Worked with members of the Institute for Artificial Intelligence in Fundamental Interactions
- Affiliated with Harvard, MIT, Northeastern, and Tufts Universities

Publications

[1] Christian Ferko, Eashan Iyer, Kasra Mossayebi, and Gregor Sanfey. *A Hodge Theory for Entanglement Cohomology*. Phys.Rev.A 111 (2025) 3, 032422. arXiv:2410.12529.

Research Experience

Algebraic K-Theory for Resources *Summer 2026*

IAIFI Visitng student researcher

- Applied algebraic k-theory, particularly Waldhausen $S_\bullet$-construction, to obtain new computable invariant for entanglement
- Applied Waldhausen $S_\bullet$-construction to obtain spectra for general resource categories of arxiv:1409.5531
- Used generalized construction on quantum magic
- Learned about and explored usage of delooping machines including Segal and May delooping machines Preprint in preparation

A_∞ algebras and Entanglement *March 2026-*

IAIFI Visiting student Researcher

- Extended entanglement DGA to an A_∞ algebra via Homotopy Transfer
- Proved Local Unitary invariance of transferred A_∞ algebra
- Extracted a variety of basis independent invariants from higher Massey products
- Preprint in preparation

Extending Generative effects to Entanglement *December 2024- ?*

with Christian Ferko

- Worked to extend the notion of generative effects of [Adams 2017] to detecting entangled states
- Learned about Von-Neumann algebras and Haag duality as a possible route

A Hodge Theory for Entanglement Cohomology *July 2024 - January 2025*

with Christian Ferko, Eashan Iyer, and Gregor Sanfey

- Built off of Mainiero's work (arxiv: 1901.02011) to apply hodge-theoretic framework to Entanglement cohomology
- Worked on proving compatibility of support operators
- Published in Physical Review A (PRA: 111.032422)

Homotopical Equivalence of Categories *March 2023 - July 2023*

Personal Project

- Furthered work done by Gurski, Johnson, and Orno (arxiv: 1508.00054) on relative categories
- Proved novel conditions necessary for adjoint equivalence of relative categories to extend to homotopical equivalence of categories
- Provided an informal talk presenting work done by Gurski et al. and my results

Talks

Mathematics Undergraduate Colloquium *University of Toronto*

GENERATIVE EFFECTS AND QUANTUM ENTANGLEMENT *October 2025*

- Presented categorical approach behind generative effects as described by [Adams 2017]
- Introduced and explained future directions for Generative effects including Segre Embedding and entanglement cohomology
- Discussed possibility of extending generative effects to descent condition for higher categories

Physics Undergraduate Colloquium *University of Toronto*

AN INTRODUCTION TO ENTANGLEMENT COHOMOLOGY *January 2026*

- Introduce entanglement of the GNS complex as described in Mainiero (arxiv: 1901.02011) and Ferko et al. (2410.12529)
- Use 2-qubit example as an explicit example of why the GNS complex accurately detects entangled states